

Reading Guide for Newton's Laws

Newton's First Law

Read p. 26 in *Conceptual Physics* then answer the following questions:

1. What is the key word in the Law of Inertia?
2. The figure below shows someone pulling a tablecloth out from under a place setting. Use the key word for the Law of Inertia to explain how this works.



Figure 2.4: Inertia in action.

3. Pick one of these figures and explain what is happening using Newton's First Law.

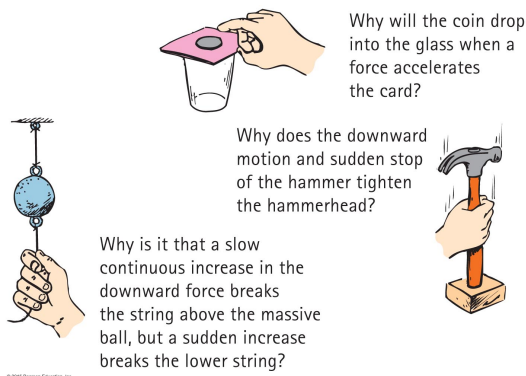


Figure 2.5: Examples of inertia.

Newton's Second Law

Read pp. 63-64 in *Conceptual Physics* then answer the following questions:

4. What does "directly proportional" mean?
5. What does "inversely proportional" mean?
6. How is acceleration related to net force and mass?

7. What is this figure trying to show?

Force of hand
accelerates
the brick



The same force
accelerates 2 bricks
 $\frac{1}{2}$ as much



3 bricks, $\frac{1}{3}$ as
much acceleration



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Figure 4.11: Acceleration is inversely proportional to mass.

Newton's Third Law

Read pp. 75-76 in *Conceptual Physics* then answer the following questions:

8. What is an *interaction*?
9. Use the concept of interaction to explain why this guy doesn't topple over.



Figure 5.2.

10. When you walk, you do so by exerting an action force with your feet pushing against the floor backwards. What is the reaction force?
11. If a car's tires push on the road as an action force, what is the reaction force?